



Fig. 4: Image with quantisation levels reduced to 64



Fig. 5: Image with quantisation levels reduced to 8

Image quantisation can be described as a mapping process by which groups of datapoints (several pixels within a range of grey values) are mapped to a single point (a single grey level). An image can be re-quantised in MATLAB using the `grayslice` function. The following set of commands reduces the quantisation levels to 64 and displays the image:

```
A = imread('Penguins_grey.jpg');
B=grayslice(A,64);
imshow(B,gray(64))
```

The resultant image of the `imshow` function is shown in Fig. 4.

The following set of commands reduces quantisation levels to 8 and displays the image:

```
A = imread('Penguins_grey.jpg');
B=grayslice(A,8);
imshow(B,gray(8))
```

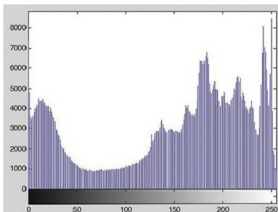


Fig. 6: Histogram of image Penguins_grey.jpg generated using the `imhist` function

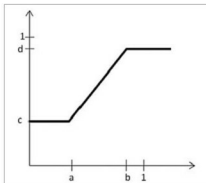


Fig. 7: Histogram stretched using `imadjust` function

The resultant image of the `imshow` function is shown in Fig. 5.

Histogram

Histogram of a greyscale image represents the frequency of its grey levels occurrence. It is a graph indicating the number of times each grey level occurs in the image. In a dark image, grey levels (and hence the histogram) are cluttered at the lower end. In a uniformly bright image, grey levels (and hence the histogram) are cluttered at the upper end. In a well contrasted image, grey levels (and hence the histogram) would be well spread out over much of the range.

In MATLAB, the histogram can be viewed using the `imhist` function. As an example, the commands that follow can be used to display the histogram of image `Penguins_grey.jpg`:

```
A = imread('Penguins_grey.jpg');
figure(1), imhist(A);
```

Fig. 6 shows the histogram generated by the above command.

The three main operations performed on a histogram include histogram stretching, histogram shrinking, and histogram sliding. All these operations are described in the following paragraphs.

Histogram stretching

This technique, also known as input cropping, consists of a linear transformation that stretches part of the original histogram so that its non-zero intensity range occupies the full dynamic grey scale.

If the histogram of the image is cluttered at the centre, it can be stretched using `imadjust` function. The following command stretches the histogram as shown in Fig. 7:

```
imadjust(F,[a,b],[c,d])
```

The values of `a`, `b`, `c` and `d` must be between 0 and 1.

Command `imadjust(F,[],[1,0])` inverts the grey value of the image, to produce a result similar to photographic negative.

Histogram stretching using above commands requires user input. An alternative approach is to use histogram equalisation command, which is entirely an automatic procedure. Histogram equalisation command in MATLAB is `histeq`. The use of `histeq` command is shown below:

```
A = imread('Penguins_grey.jpg');
HE = histeq(A);
imshow(HE),figure, imhist(HE)
```

Figs 8 and 9 show the image and the equalised histogram, respectively.

Histogram shrinking

This technique, also known as output cropping, modifies the original histogram such that its dynamic greyscale range is compressed into a narrower greyscale. `imadjust`